

Introduction of battery charging box--T-BAT-Charger



Written by Jay Luo. 10.05.2024

Overview

The SolaX T-BAT-Charger is an accessory designed for charging the battery to the desired state of charge (SOC) when it cannot be charged using conventional methods. The T-BAT-Charger has been integrated with functions for the battery status checking and power supply for charging, which allow it to realize both checking and charging the battery.

Appearance

The figures below show the basic appearance and dimensions of the T-BAT-Charger.



Figure 1. T-BAT-Charger

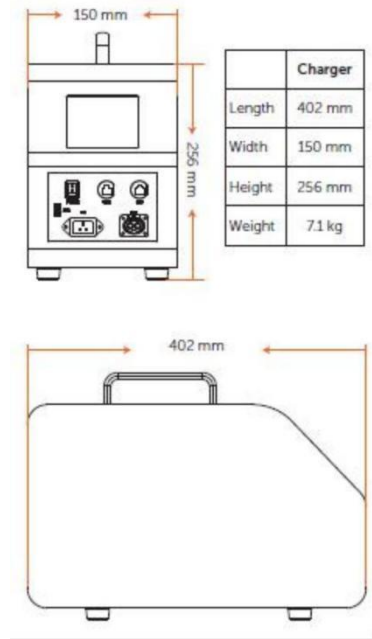


Figure 2. Dimensions of the T-BAT-Charger

Features

Thanks to the compact and excellent industrial design, the T-BAT-Charger has the following features:

- Easy to carry, easy to use;
- Support up to 2 PCS batteries;
- High charging efficiency up to 90%;
- Smart Cooling system, 0 °C ~50 °C operation temperature supported.

Technical Data

The table below shows the technical data of the T-BAT-Charger.

Table 1. Technical Data Sheet

Electrical Characteristics	
Max. AC Input Apparent Power	2200 VA
Max. AC Input Current	10 A
Nominal AC Voltage (AC Voltage Range)	220 V (180~264 V)
Nominal Grid Frequency Range	47-63 Hz
Efficiency	90%
Surge Current (230 V Cold Start)	60 A
Output Voltage Range	40-150 V
Charging Current Range	0-12 A
Voltage Accuracy	1.0%
Max. Number of Batteries Supported	2 PCS
Operating Conditions	
Enclosure Protection	IP20
Operating Temperature Range	0°C to 50°C
Relative Humidity Range	20%RH to 90%RH (Non-condensing)
Storage Temperature Range	-40°C to 85°C
Storage Humidity Range	10%RH to 95%RH (Non-condensing)
Cooling Type	Smart Cooling

Ports Definition

Generally, the interfaces of the T-BAT-Charger can be divided into three parts according to its functions. The power supply ports, the communication ports and the upgrading ports.

The figure below shows the exact definition of each port.

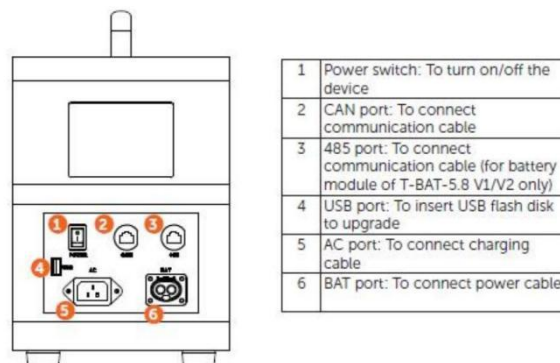


Figure 3. Interfaces of the T-BAT-Charger

Power Cable Introduction

Each T-BAT-Charger comes with a power cable, which consists up to 5 pairs of terminals and makes it compatible with multiple battery types. The figure below shows the diagram and a general introduction of the power cable.

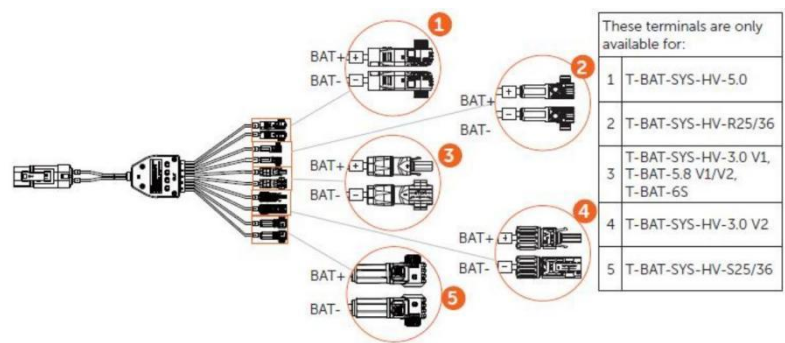


Figure 4. Power Cable

Besides, there are 8 clips (4 for each sides) on this cable to secure the unused lines. The figure below illustrates the diagram of clips.

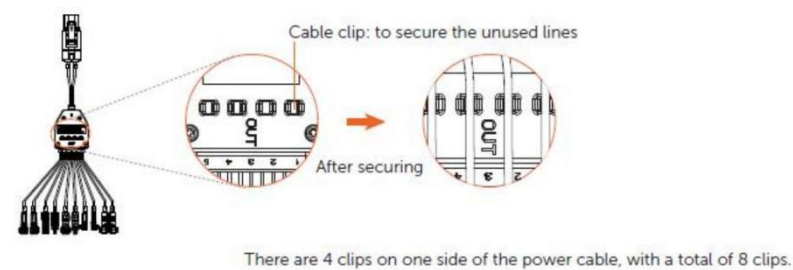


Figure 5. Cable Clip

Wiring

Since there are several different SolaX battery series and each may have different ports and terminals design with each other, there might be some slight difference on the wiring methods according to the battery types.

In this section, the wiring diagrams with different battery types are given as following pages, please refer to the relevant section according to the battery model.

Connectable Battery Models

T-BAT-SYS-HV-3.0

T-BAT-SYS-HV-5.0

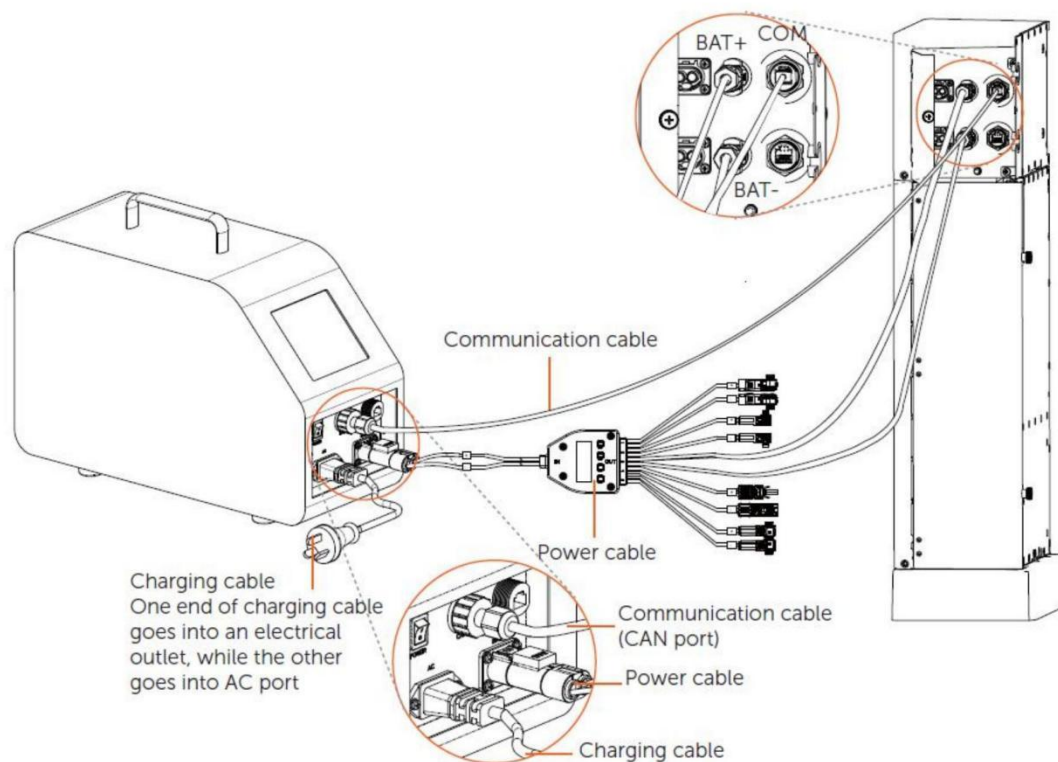
T-BAT-5.8 V1/V2

T-BAT-SYS-HV-R25/36

T-BAT-SYS-HV-S25/36

T-BAT-6S

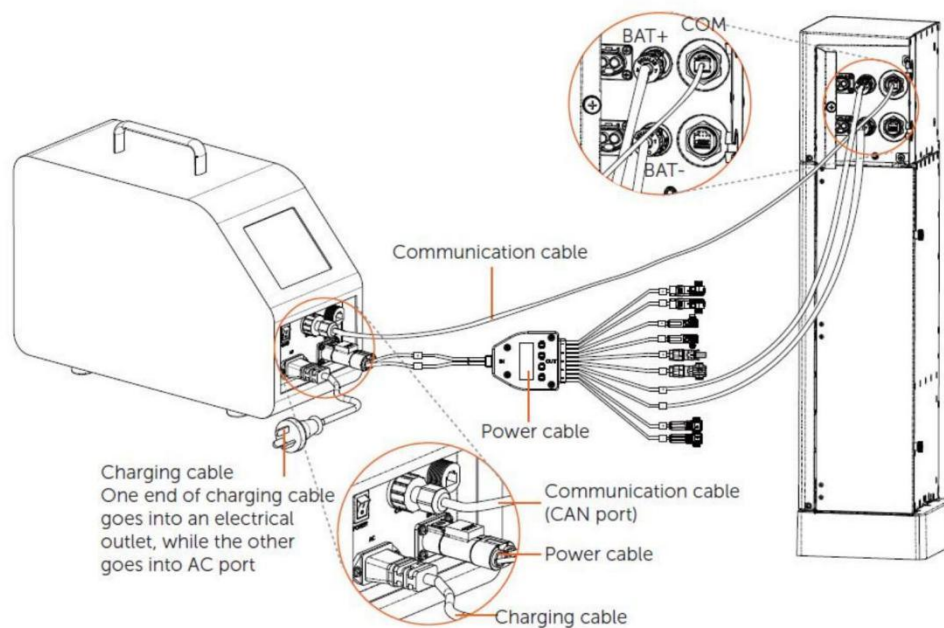
T-BAT-SYS-HV-3.0 V1



***Note:**

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Only **one** battery can be charged at once with the Charger.

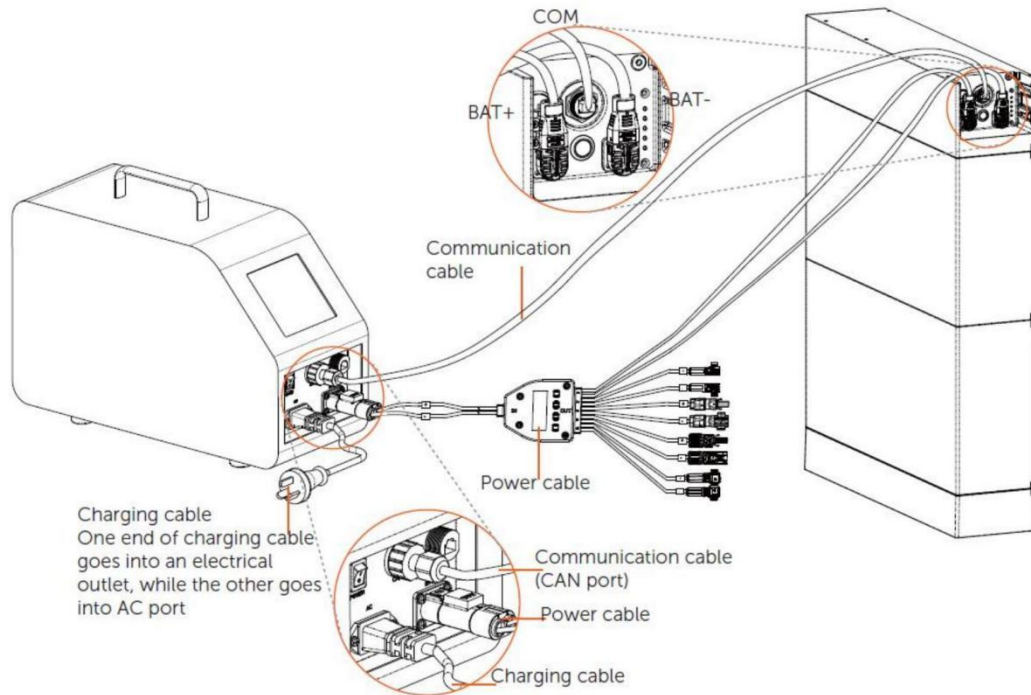
T-BAT-SYS-HV-3.0 V2



*Note:

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Only **one** battery can be charged at once with the Charger.

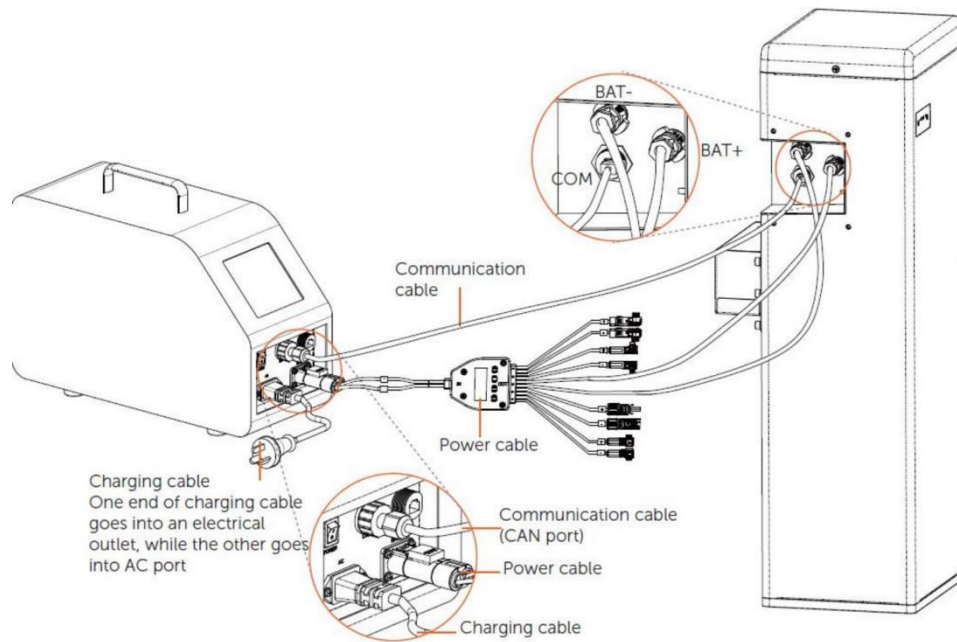
T-BAT-SYS-HV-5.0



*Note:

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Up to **two** batteries can be charged at once with the Charger.

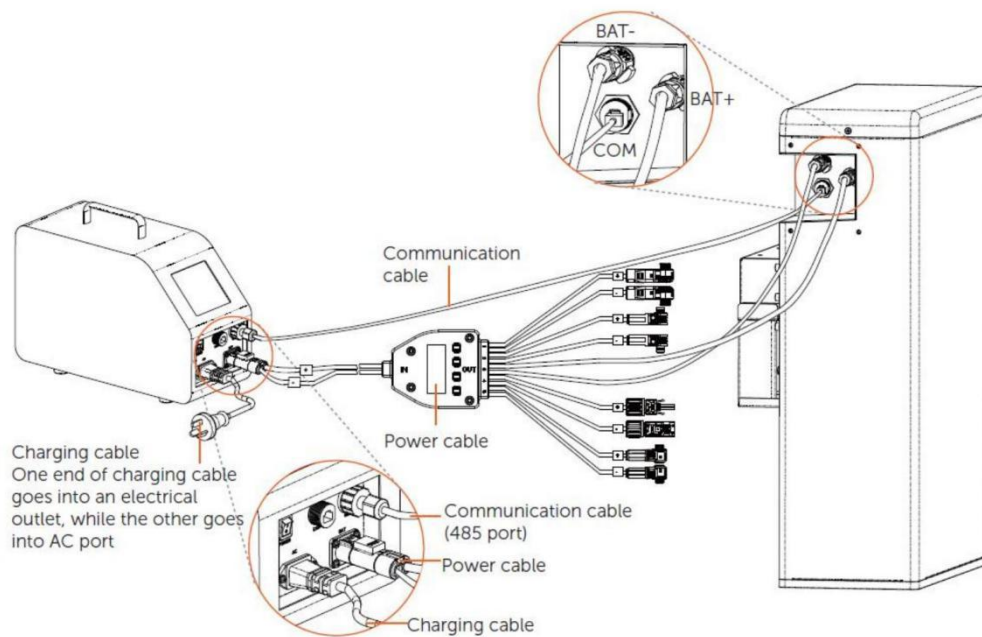
T-BAT-5.8 V1/V2 (with BMS)



*Note:

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Only **one** battery (T-BAT-5.8 V1/V2 with BMS) can be charged at once with the Charger.

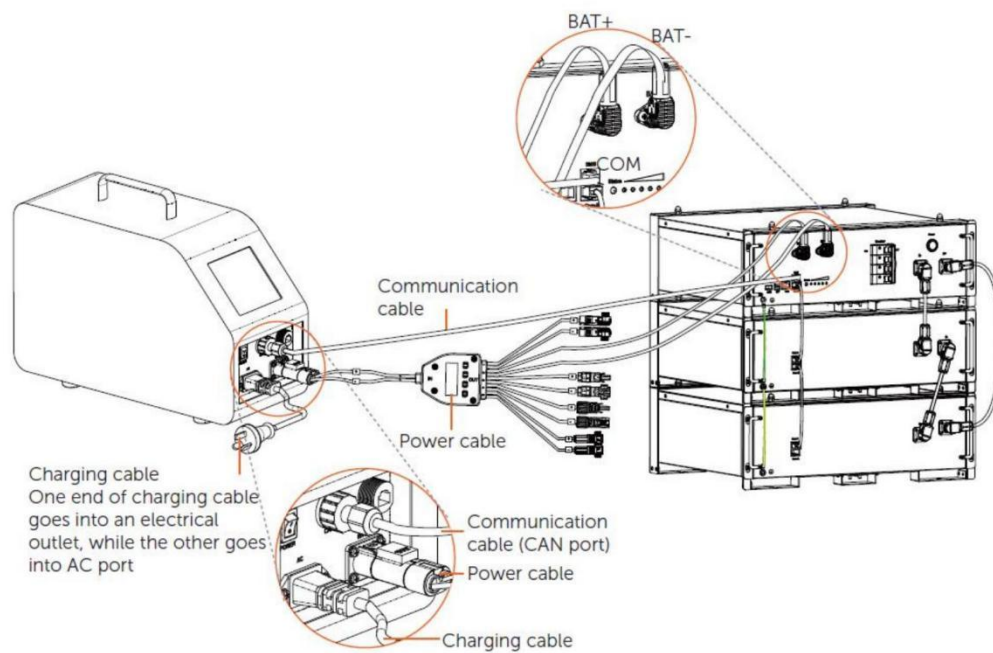
T-BAT-5.8 V1/V2 (without BMS)



*Note:

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Only **one** battery (T-BAT-5.8 V1/V2 without BMS) can be charged at once with the Charger.

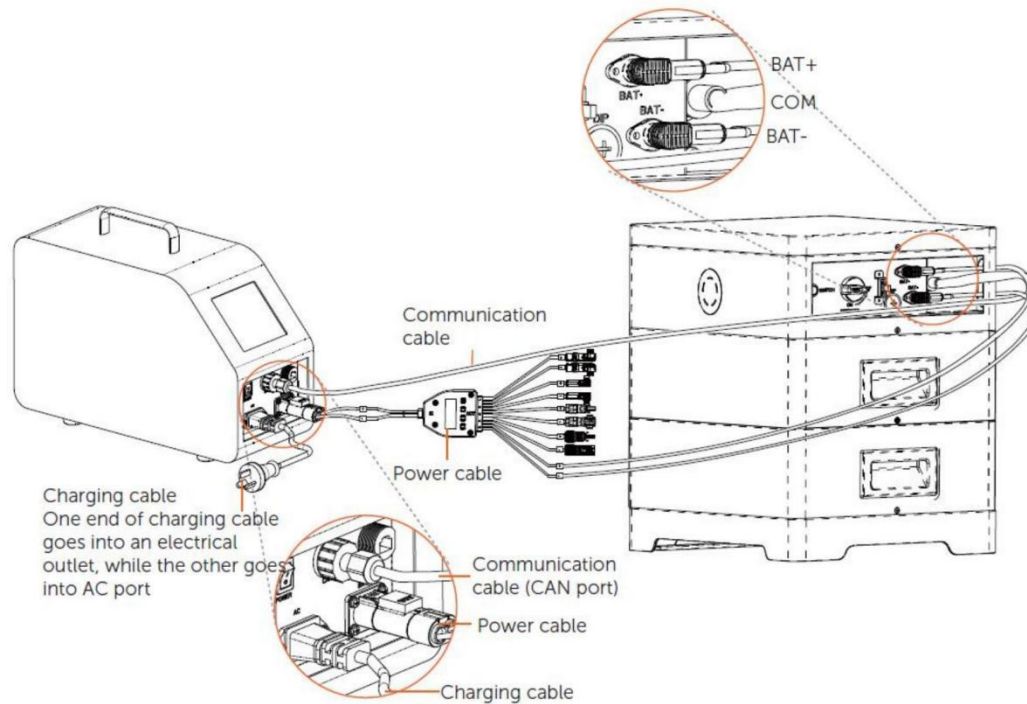
T-BAT-SYS-HV-R25/36



***Note:**

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Up to **two** batteries can be charged at once with the Charger.

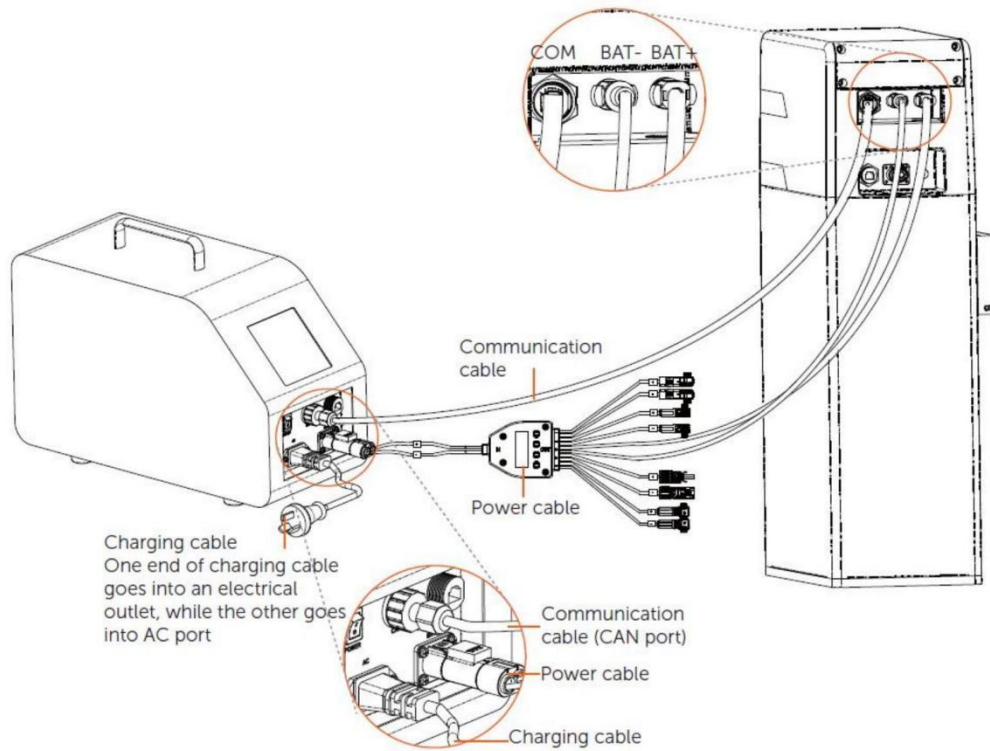
T-BAT-SYS-HV-S25/36



*Note:

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Up to **two** batteries can be charged at once with the Charger.

T-BAT-6S

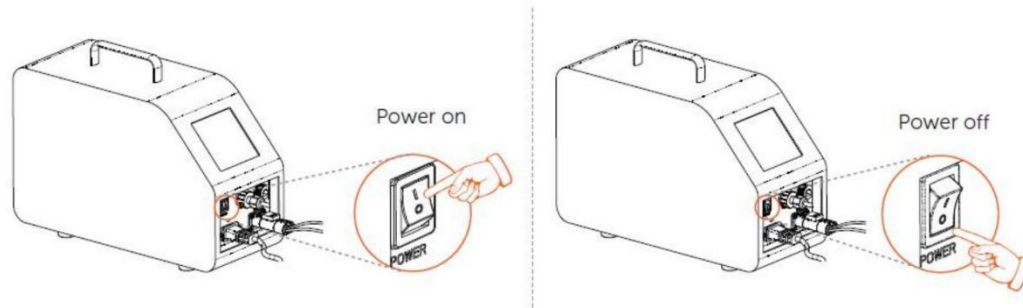


***Note:**

1. The remaining connectors on the power cable must be snapped into the cable clips. There are 4 clips on one side of the power cable, with a total of 8 clips.
2. Connectors must be protected from short-circuiting, immersion in water and moisture. Users should avoid coming into contact with the conductor materials.
3. Only **one** battery can be charged at once with the Charger.

Power On

After finish the wiring operation, toggle the “POWER” switch to start the charger.



Screen Introduction and Operation

The T-BAT-Charger is equipped with an LCD screen for information display and device control. In this section, the detailed introduction and operation of the screen is provided.

Menu Introduction

The figure below shows the menu interface with definitions of each sections noted. The status of the battery can be obtained on the main page. Besides, the charging settings can be set by filling the charging parameters (Charge SOC and Charge Current). Tapping the icons of settings and Details refers to the T-BAT-Charger setting page and the detailed information of the battery cells.

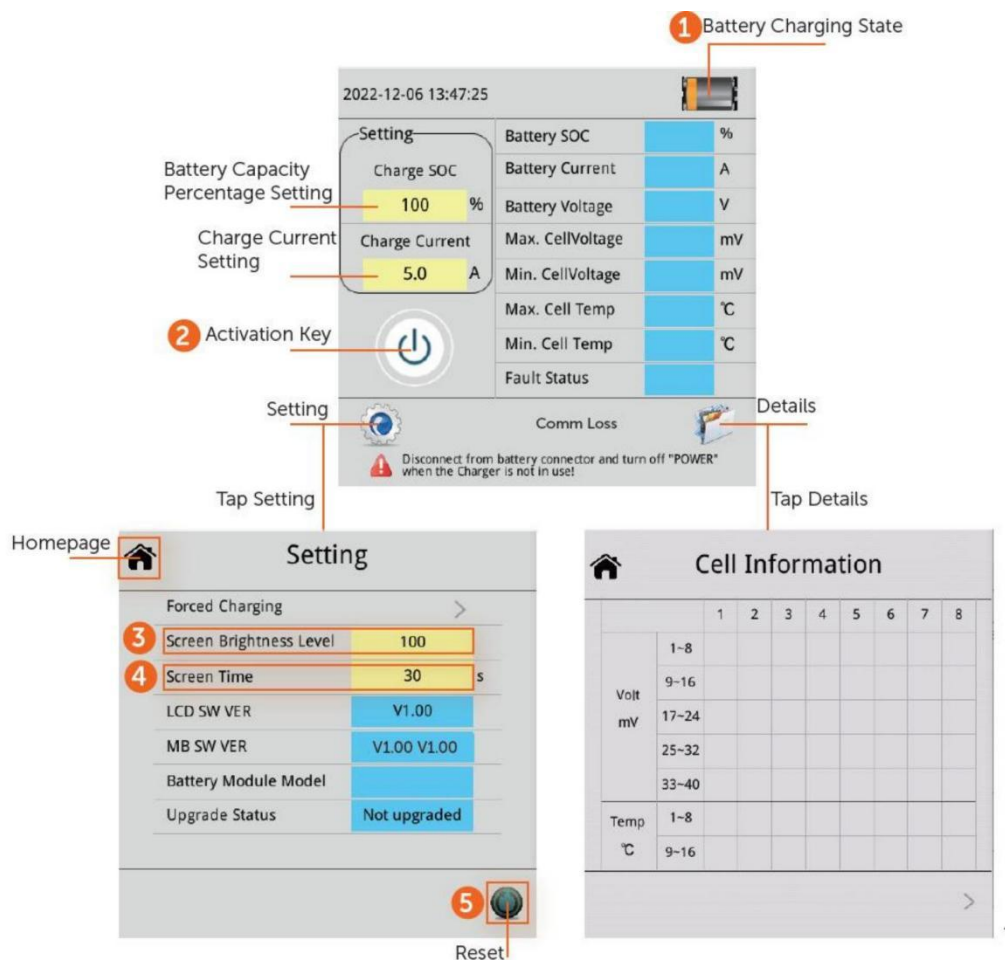


Figure 6. Menu

*Note:

1. Battery Charging State: If the icon is green and the green light cycles, the device is charging. Otherwise, it is not charging.
2. Activation Key: If the Key snaps and the colour shows white, the device will stop charging; if the Key is pressed and the colour shows yellow, the device will start charging.
3. Screen Brightness Level: The screen brightness can be manually adjusted by the user, with a level of 0 to 100.
4. Screen Time: The screen time also can be adjusted by the user, with a time range of 30 to 240 seconds.

Written by Jay Luo. 10.05.2024

Forced Charging

The Charger supports the function of "forced charging," and the function requires a password to start.

1. Tap the "Setting", and the user will see the "Forced Charging";
2. Tap the "Forced Charging";
3. Enter a password;

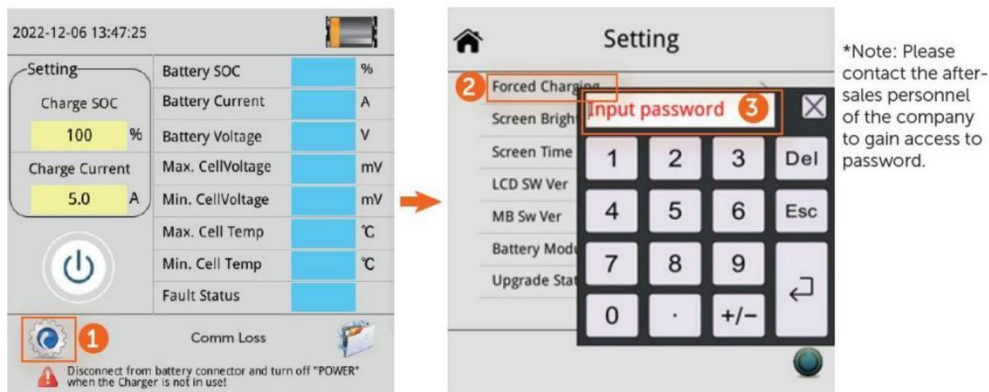


Figure 7. Forced Charging (1)

4. Tap to set a time limit for 5 to 15 minutes before charging. The default time limit is 5 minutes;

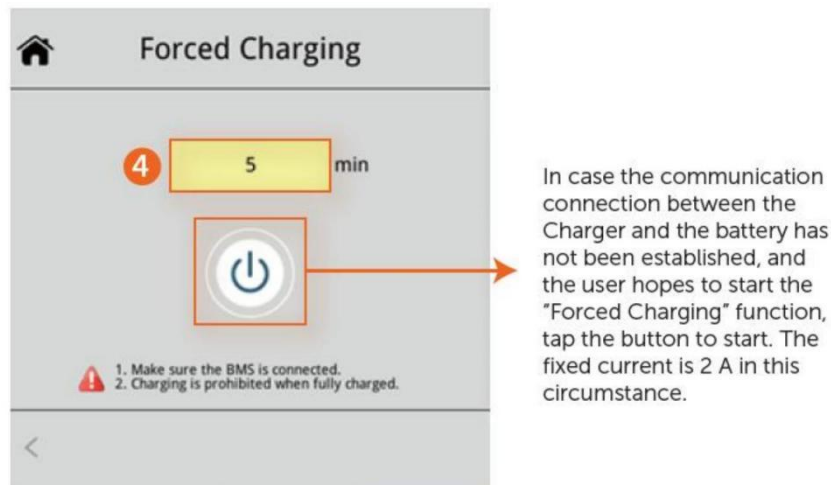


Figure 8. Forced Charging (2)

*Note: The "Forced Charging" must be disabled under any of the following

Written by Jay Luo. 10.05.2024

circumstances:

1. The ambient temperature is below 0°C or over 55°C;
2. The SOC of the battery reaches 100%.

Under the following circumstances, the "Forced Charging" will be stopped:

1. Time's up;
2. Battery fault;
3. Communication between the Charger and battery is established;
4. Tap the stop button.

Upgrade

The Charger supports the function "Upgrade". Please follow the steps below to upgrade the device:

1. Contact the SolaX post-sales staff to obtain the Upgrade File;
2. Transfer the file into a USB Flash Disk (USB 2.0 or later is required);
3. Insert the USB Flash Disk, tap "Yes" to jump to "Reset" page. The upgrading process will start automatically.

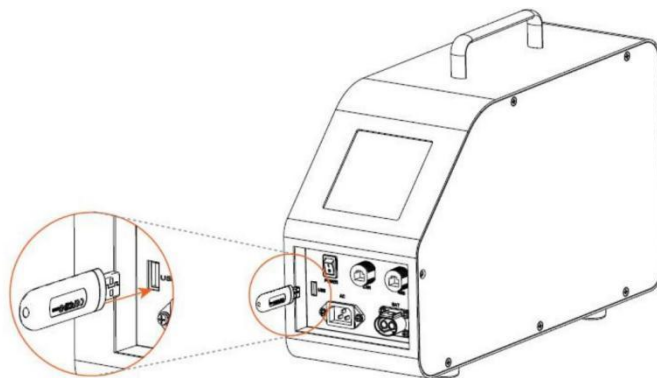


Figure 9. USB upgrade

*Note: The USB flash disk will not be provided by the company.

The figure below illustrates the upgrading process.

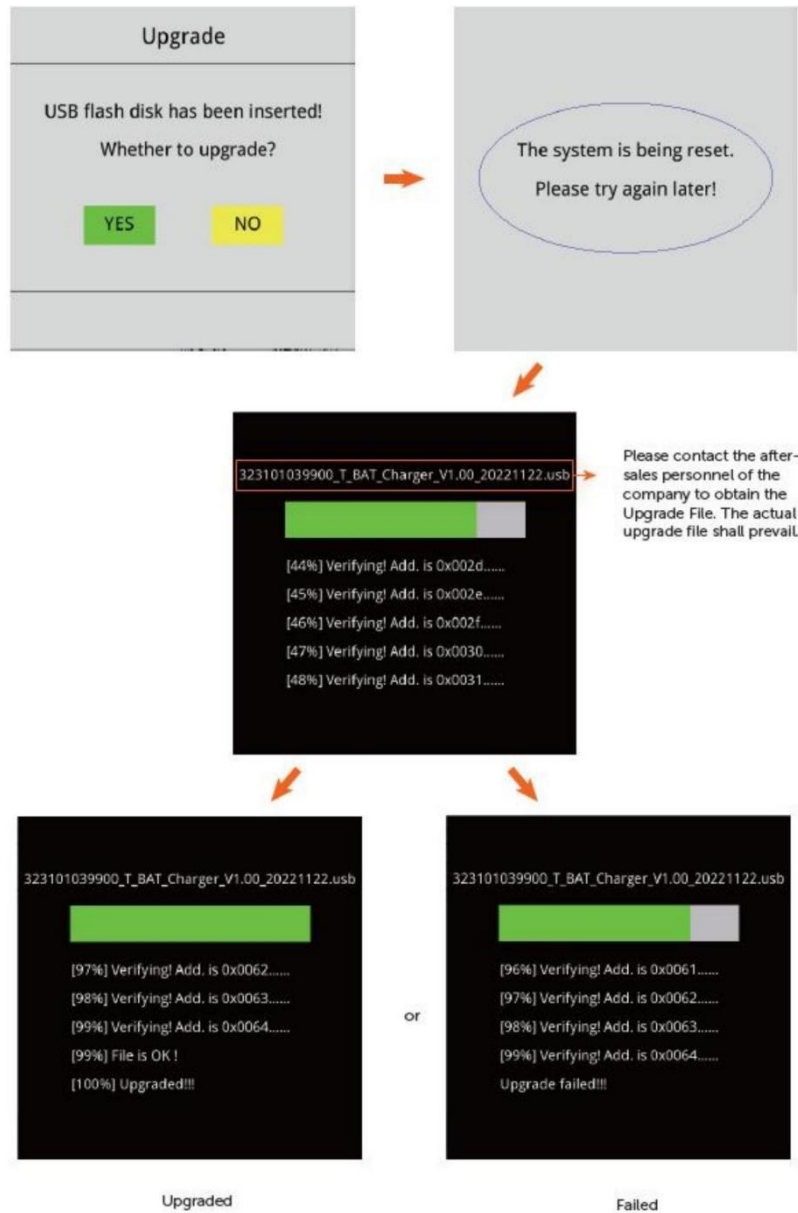


Figure 10. Upgrading Process

*Note: Please DON'T shut down the device, DON'T remove the USB flash disk, and DO keep it being plugged in all the time while upgrading, or the upgrade will fail.

Written by Jay Luo. 10.05.2024

